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Disruption is the new normal

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In 2022, the world found itself in the midst of a global energy crisis spurred on by war in Ukraine and increasingly frequent and severe climate events.

In many places, the situation looked dire. In the UK, [electricity prices rose by 65.5%](#) and gas prices by 128.9% in the 12 months to November 2022. Many European countries feared a colder-than-usual winter combined with the absence of Russian gas supply could further exacerbate an already struggling energy ecosystem. Add the lingering shadows of Covid-19 and its variants and it's not surprising that few feel optimistic about what 2023 may bring.

Let's take a look at what energy sector changemakers will be focusing on this year.

Domestic energy resiliency

In the face of volatile market behaviour, countries will increasingly focus on bolstering energy resiliency within their own borders. Across Europe, policy has been used to manage the demand/supply imbalance, including managing street lights and illuminated signage. The UK and Poland have introduced financial incentives for consumers in a bid to save energy.

In 2023, the focus must turn to long-term grid resiliency solutions, such as domestic renewable energy generation and bolstering national grid efficiency. [According to the Australian Energy Market Operator's Step Change scenario](#), improved grid efficiency may not only result in greater energy security, but also a reduction in electricity pricing by as much as 30-40%.

Distributed Energy Resources (DERs), combined with decentralised storage and AI offer a multi-beneficial way to achieve this. The groundwork has already been laid, but we can expect increased heavy investment in this technology. In time, decentralised energy systems will become an indispensable part of the grid.

Prosumer tech

The mass rollout of prosumer technology, such as home solar generation and battery storage, is one gateway to making DERs a reality.

Prosumer technology is quickly becoming more efficient and less costly, and improvements in AI are making the benefits more real for consumers. We're moving toward a world where connected tech will enable peer-to-peer microgrids and energy trading. AI will accelerate efforts to serve smart cities with this technology, going so far as to offer two-way charging stations via electric vehicles. As partners in the energy transition, consumers and policymakers alike will be interested not only in energy security within national borders, but also within the home.

Shift to renewables

The northern hemisphere droughts, heatwaves and war in Ukraine forced countries to revert to coal and nuclear-fired generation in 2022 to meet demand. While Germany delayed its planned closure of nuclear plants in 2022 in the face of energy security challenges, while Japan is planning to restart seven nuclear power plants in 2023. With meteorologists predicting a rare third consecutive year of La Niña weather patterns in 2023, we should expect more of the same.

In spite of these short-term measures to meet sudden and critical demand, renewables are on the rise. Global solar capacity is expected to [surpass a terawatt of power generation](#) in 2023. According to the International Energy Agency, 33% of global electricity is forecast to be renewable by 2024, with solar accounting for almost 60%. As this technology develops, it's becoming more efficient and cost-effective - the reduction in the cost of solar is expected to retain its downward trajectory, having [reduced by 85% in the past decade](#).

[Green hydrogen is also on the horizon](#), predicted to be worth \$201 billion by 2025, up from \$130 billion in 2020. [The global green hydrogen pipeline](#) currently exceeds 250GW - 200 times the power produced in 2020 - with the first gigawatt-scale plant anticipated to fire up in 2028.

Increased legislation

As 2030 nears, clean and affordable energy targets are looming larger. Legislation from institutions like the UN, IRENA and government coalitions is putting pressure on organisations to transition to the green energy future.

In the US, 22 states plus Washington DC have set up enabling policies for community solar. The Biden administration has offered a solar investment tax credit, and private sector investment in renewables is expected to explode in 2023. We can expect to see government policies that directly shape the energy industry and its outcomes toward a greener future in most developed countries.

Partner collaboration

We are part of an energy ecosystem, and not one of us can meet 2030 targets, enable the energy transition, or establish domestic energy security alone. Take AI as an example. The technology exists to realise the potential role of this game-changing development, but AI-supporting business practices need to be adopted in order for this to happen.

The collaboration doesn't stop at business. On a different scale, we need engagement from consumers. DER, home battery storage, EVs and AI make consumers our partners in the energy transition - we need their buy-in. Partner collaboration was a strong theme of the 2022 Enlit conference in Europe, so we can expect a strong focus on collaboration in 2023 and beyond.

With war, energy scarcity, climate catastrophe and economic crisis all casting dark shadows around the globe, it can be hard to look positively on the macro environment of the energy industry as we approach 2023. But the interplay of these five factors - domestic resiliency, prosumer tech, the shift to renewables, increased legislation and partner collaboration - represents a tipping point for the global energy industry as we know it. We're on the brink of a cleaner, more cost-effective and more resilient energy future, and perhaps the climate we're in right now will serve as the catalyst.

For energy retailers, the role you play in this tricky, eternally disrupted landscape is vital. Your work shepherds consumers through the energy transition. The best thing you can do to keep ahead of the game and support the critical evolution of the industry is to digitise your systems now. Shall we collaborate?